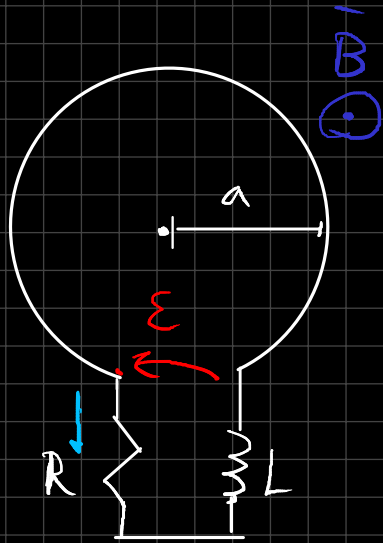


Ejercicio 4

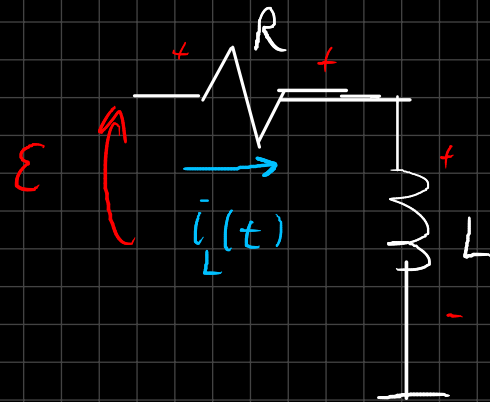
Se tiene una espira circular de radio a , resistencia R y autoinductancia L , perpendicular a un campo de inducción $B = B_0$. Si el campo se apaga exponencialmente, o sea, $B(t) = B_0 e^{-t/\tau}$, donde τ es una constante de tiempo, calcule la corriente inducida en la espira.

También lo
empezamos en
la práctica
del 27/8



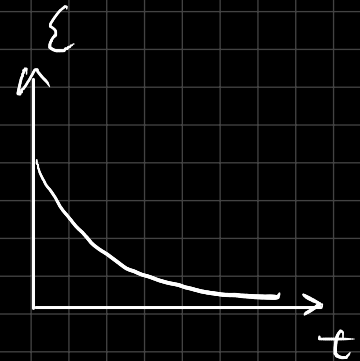
$$\phi = \pi \cdot a^2 \cdot B(t) = \pi \cdot a^2 \cdot B_0 e^{-t/\tau}$$

$$\varepsilon = -\frac{d\phi}{dt} = \frac{\pi \cdot a^2 \cdot B_0}{\tau} e^{-t/\tau} \cdot u(t)$$



Kirchhoff:

$$\varepsilon - R \cdot i(t) - L \cdot \frac{di}{dt} = 0$$



$$\Rightarrow \text{LAPLACE: } E(s) - R I(s) - L \cdot s \cdot I(s) = 0 \Rightarrow E(s) = (R + sL) I(s)$$

$$\Rightarrow I(s) = \frac{1}{R + sL} \cdot E(s) = \frac{1/L}{s + R/L} E(s)$$

$$E(s) = \mathcal{L} \left\{ \frac{\pi \cdot a^2 \cdot B_0}{\tau} e^{-t/\tau} \cdot u(t) \right\} = \frac{\pi \cdot a^2 \cdot B_0}{\tau} \cdot \frac{1}{s + 1/\tau}$$

$$\Rightarrow I(s) = \frac{1/L}{s + R/L} \cdot \frac{\pi \cdot a^2 \cdot B_0}{\tau} \cdot \frac{1}{s + 1/\tau} = \frac{\pi \cdot a^2 \cdot B_0}{L \cdot \tau} \left\{ \frac{1}{s + R/L} \cdot \frac{1}{s + 1/\tau} \right\}$$

Acá asumí q' $R/L \neq 1/\tau \Rightarrow \tau - L/R \neq 0$

$$\frac{1}{s + R/L} \cdot \frac{1}{s + 1/\tau} \stackrel{\uparrow}{=} \frac{A}{s + R/L} + \frac{B}{s + 1/\tau} \Rightarrow 1 = A(s + 1/\tau) + B(s + R/L)$$

$$\text{Si } s = -1/\tau \Rightarrow 1 = A \cdot 0 + B(-1/\tau + R/L) \Rightarrow B = \frac{1}{R/L - 1/\tau} = \frac{L \cdot \tau}{\tau \cdot R - L}$$

$$\text{Si } s = -R/L \Rightarrow 1 = A(-R/L + 1/\tau) + B \cdot 0 \Rightarrow A = \frac{1}{1/\tau - R/L} = \frac{-1}{R/L - 1/\tau} = -B =$$

$$= - \frac{L \cdot \tau}{\tau \cdot R - L}$$

$$\Rightarrow I(s) = \frac{\pi \cdot a^2 \cdot B_0}{\cancel{L} \cdot \tau} \left\{ \frac{-\cancel{L} \tau}{\tau \cdot R - L} \cdot \frac{1}{s + R/L} + \frac{\cancel{L} \tau}{\tau \cdot R - L} \cdot \frac{1}{s + 1/\tau} \right\} = \frac{\pi \cdot a^2 \cdot B_0}{\tau \cdot R - L} \left[\frac{-1}{s + R/L} + \frac{1}{s + 1/\tau} \right]$$

$$\Rightarrow i_L(t) = \frac{\pi a^2 \cdot B_0}{R} \frac{1}{\tau - \frac{L}{R}} \left\{ -e^{-\frac{R}{L} \cdot t} + e^{-t/\tau} \right\}$$

¿Respuesta sobre amortiguada?

$$\frac{R}{L} = 1/\tau$$

$$\tau = L/R$$

$$H = \Omega \cdot J$$

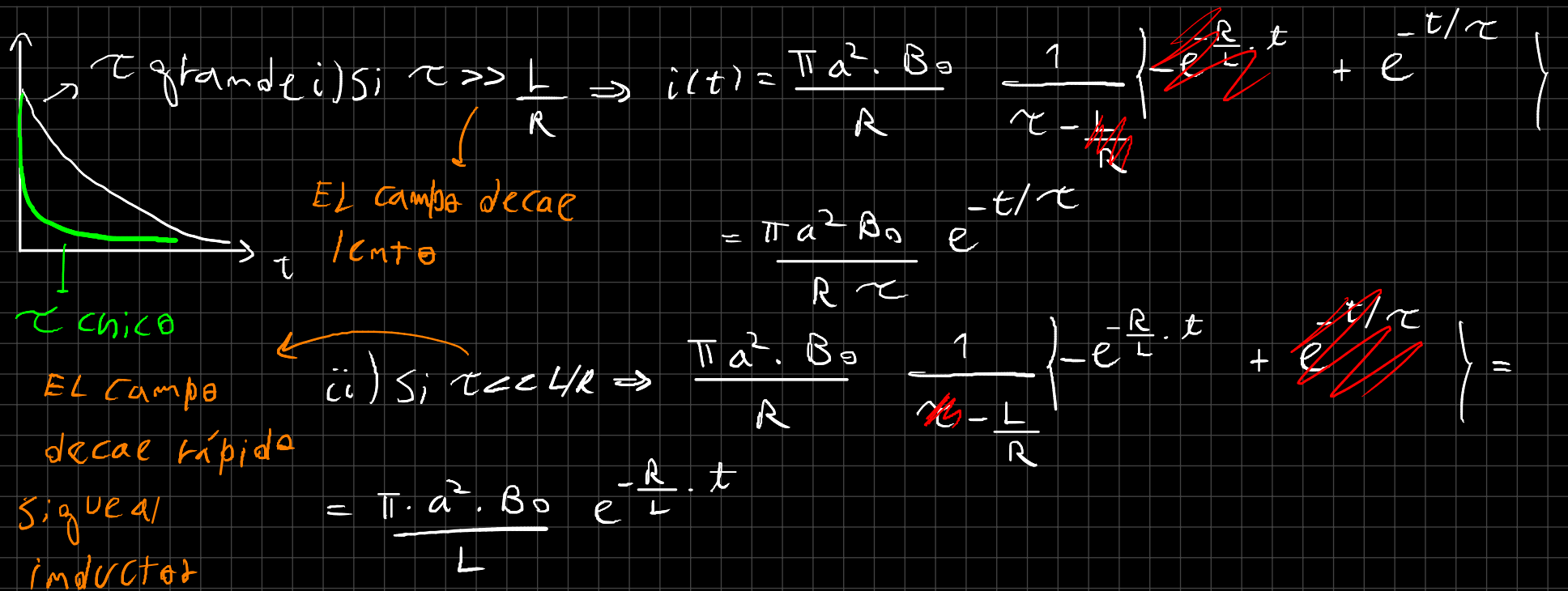
¿Qué pasa si $\tau = L/R \Rightarrow \tau - L/R = 0$?

$$\rightarrow \frac{1}{s + R/L} \cdot \frac{1}{s + 1/\tau} = \frac{1}{(s + 1/\tau)^2} \xrightarrow{\mathcal{L}^{-1}} t e^{-t/\tau}$$

$$\Rightarrow i(t) = \frac{\pi \cdot a^2 \cdot B_0}{L \cdot \tau} \cdot t e^{-t/\tau} = \frac{\pi \cdot a^2 \cdot B_0}{L} \left(\frac{t}{\tau} \right) e^{-t/\tau}$$

¿Críticamente amortiguada?

¿Por qué no hay subamortiguada? $\rightarrow$ xq' los polos deben ser reales



iii) $\tau = L/R \Rightarrow$ Críticamente amortiguada

Loco.